HANDSMEN THREADS

Salesforce CRM Implementation

Project Documentation

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1. PROJECT OVERVIEW

HandsMen Threads is a Salesforce-based Customer Relationship Management solution developed to organize and streamline important business activities within a fashion and apparel business environment. The project provides a centralized platform for managing business information and supporting processes related to customers, products, orders, inventory and other relevant CRM activities.

The solution combines Salesforce configuration, automation and Apex development to improve data consistency and reduce repetitive manual activities. Custom Salesforce components are used to structure business information, while validation and automation features help ensure that business processes are performed consistently.

The project also includes Apex-based business logic for order quantity validation and low-stock inventory monitoring. Reports and dashboards provide a summarized view of business information, while Salesforce security features help control access according to user responsibilities.

Overall, HandsMen Threads demonstrates how Salesforce can be customized to provide an integrated CRM solution that supports business operations, improves information visibility and provides a foundation for future enhancements.

2. OBJECTIVES

The main objective of the HandsMen Threads project is to develop a centralized Salesforce CRM solution that supports efficient management of business information and improves the consistency of important business processes.

The specific objectives are:

1. To organize business information using a structured Salesforce data model.
2. To provide centralized management of customer, product and order-related information.
3. To improve data accuracy through appropriate validation mechanisms.
4. To automate repetitive business processes using Salesforce automation.

5. To implement custom Apex logic for order quantity validation and low-stock inventory monitoring.
6. To provide useful business insights through reports and dashboards.
7. To implement appropriate user access and security controls.
8. To test the implemented Salesforce features using suitable test data.
9. To provide a scalable foundation for future CRM improvements.

REQUIREMENT ANALYSIS & PLANNING

3.1 Understanding Business Requirements

HandsMen Threads requires a centralized CRM solution to organize and manage important business information within a single Salesforce environment. The system is designed to support the management of customers, products, orders and inventory-related activities while reducing dependence on manual processes.

The CRM should allow authorized users to create and maintain business records, ensure that important information follows defined validation rules, and automate repetitive activities wherever required. The solution should also provide visibility into business information through reports and dashboards.

For processes that require customized business logic, Apex development is used. In the HandsMen Threads implementation, Apex triggers support stock detection and order quantity validation. Security configurations are used to provide appropriate access to users according to their responsibilities.

3.2 Project Scope

The scope of the HandsMen Threads Salesforce CRM includes the configuration and development of a Salesforce environment for managing the core business activities represented in the project.

The implementation covers:

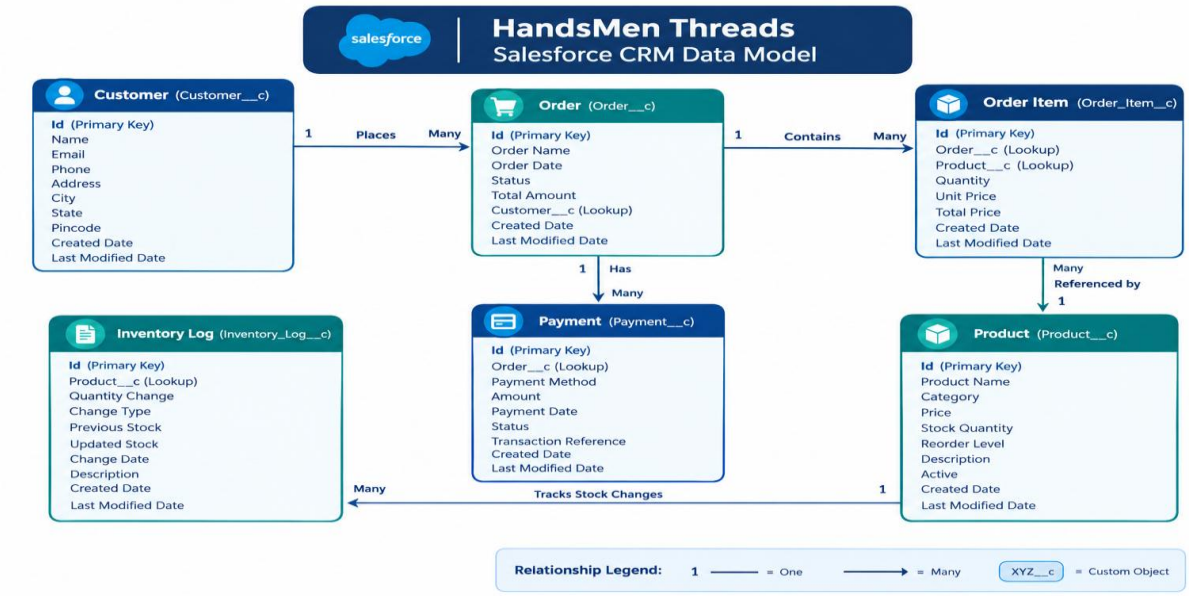
- Management of relevant business records using Salesforce objects.
- Configuration of fields and relationships between related records.

- Validation of user-entered information.
- Automation of business processes using Salesforce automation features.
- Implementation of Apex triggers for order quantity validation and low-stock inventory monitoring.
- Configuration of the HandsMen Threads Lightning application and user interface.
- Generation of reports and dashboards for business visibility.
- Configuration of user access and security controls.
- Testing of the implemented functionality using suitable test data.
- Documentation of the implemented solution and its results.

3.3 Data Model

The HandsMen Threads CRM uses Salesforce objects to represent the major business entities involved in the application. Fields are configured within these objects to capture the information required by the business process. Relationships are used where records need to be associated with related business information.

The data model is designed to keep information structured and reusable while supporting automation, reporting and security requirements.

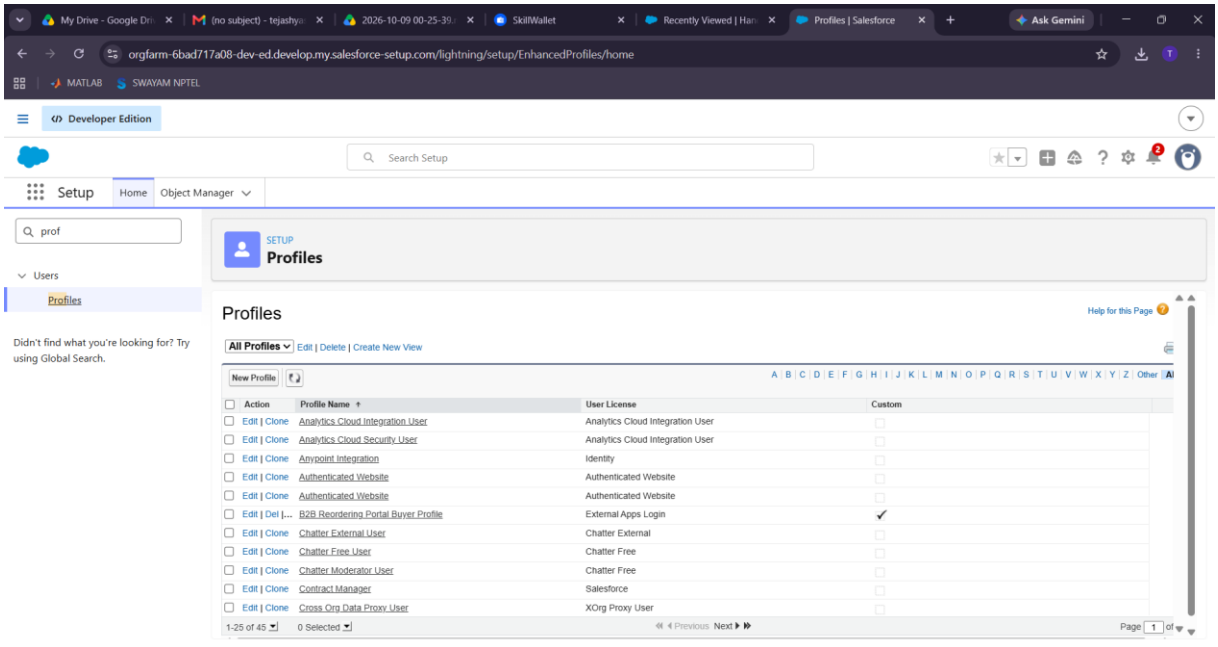


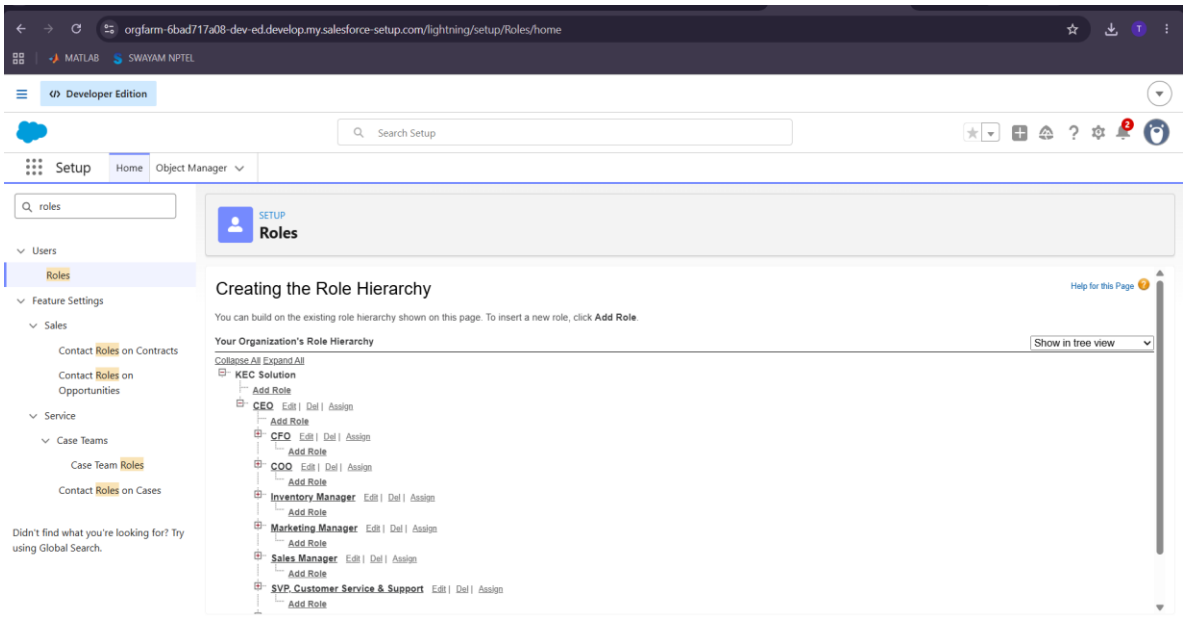
3.4 Security Model

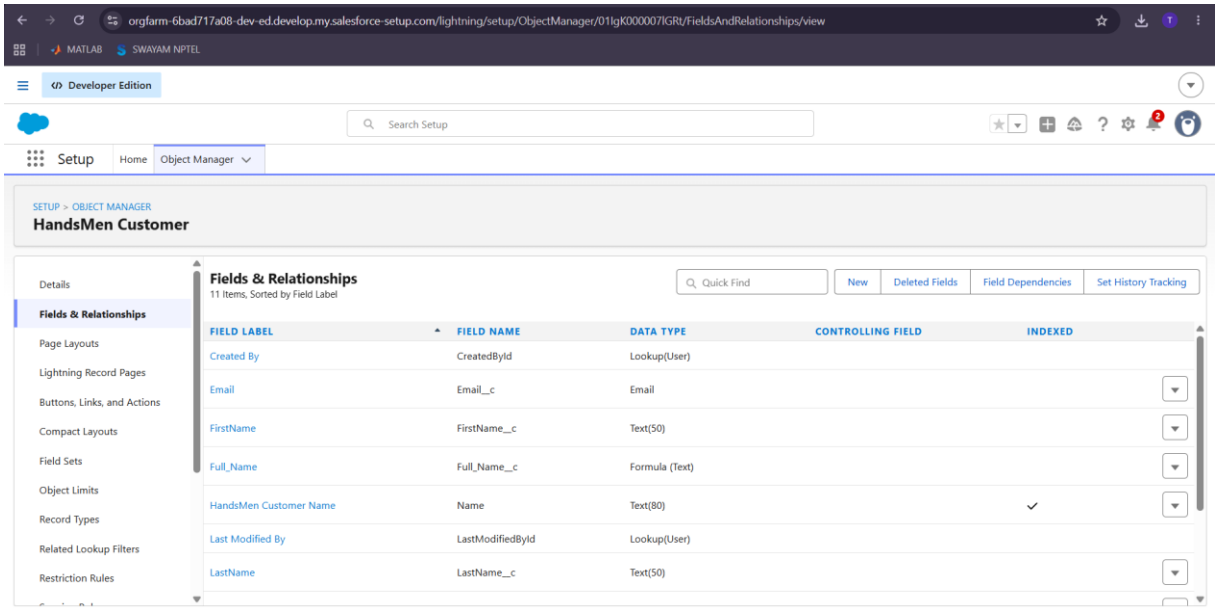
The HandsMen Threads CRM uses Salesforce security mechanisms to control access to business information according to user responsibilities. The security model is designed to provide users with the permissions required for their work while protecting sensitive business data from unauthorized access.

The security configuration includes profiles as the basic level of user access and permission sets for providing additional permissions to selected users. Roles and role hierarchy can be used to establish the organizational access structure, while sharing mechanisms control access to records when required. Field-level security is used to control whether specific fields can be viewed or edited by users.

These security mechanisms provide multiple levels of access control and help maintain the confidentiality and integrity of the information stored in the HandsMen Threads CRM.



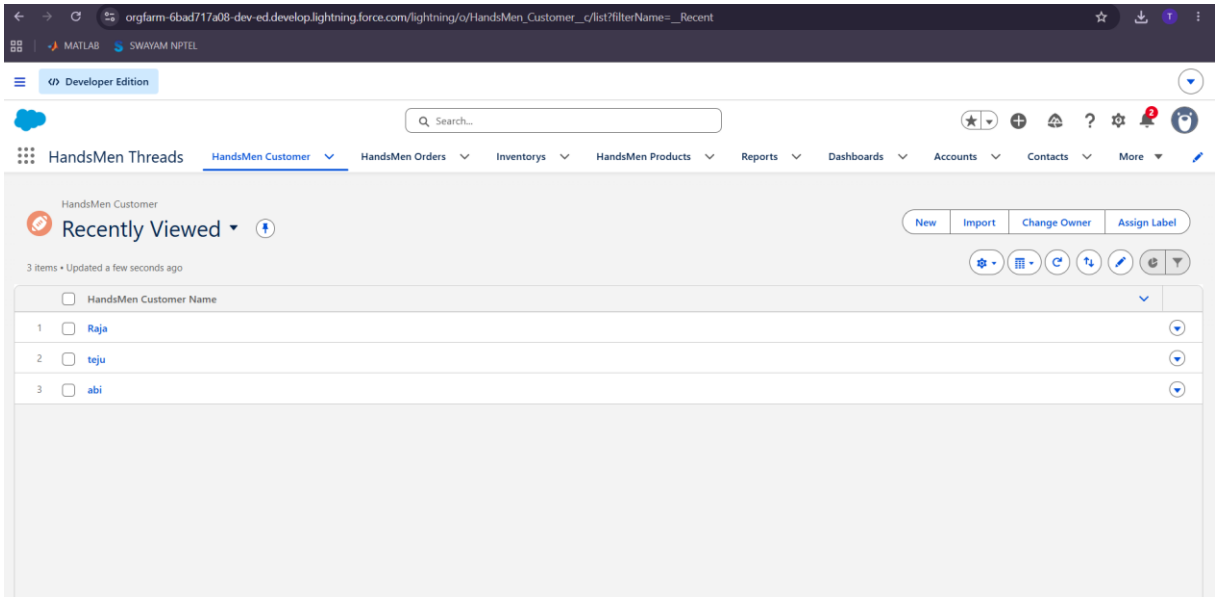




4.1 Salesforce Development Environment

The HandsMen Threads solution was developed within the Salesforce platform using Salesforce configuration tools and Apex development capabilities. Configuration-based features were used where the requirement could be addressed through standard Salesforce functionality, while Apex was used for business logic requiring programmatic processing.

The development approach focused on maintaining a clear separation between data configuration, automation, user access, and custom code. This makes the application easier to understand, test, and maintain.



4.2 Objects and Fields

Salesforce objects form the primary data-storage structure of the HandsMen Threads CRM. Each object represents a defined business entity, while its fields capture the attributes required for that entity.

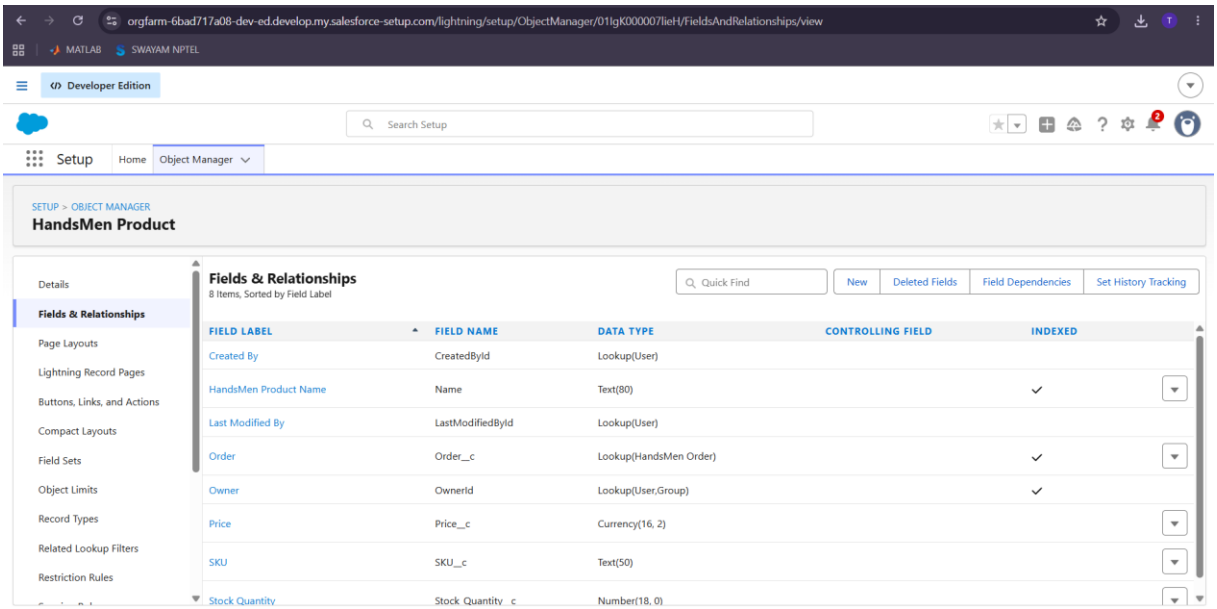
The fields were selected according to the information needed by the application's business processes. Appropriate Salesforce data types are used to represent values such as text, dates, numbers, currency, selections, and relationships.

This structured approach improves data consistency and provides reliable information for automation and reporting.

4.3 Object Relationships

Relationships connect records belonging to different business entities and allow Salesforce to maintain the associations required by the application.

In HandsMen Threads, relationship fields are used wherever one business record depends on information maintained in another record. These relationships support navigation between related records and allow automation and reporting to work with connected information.

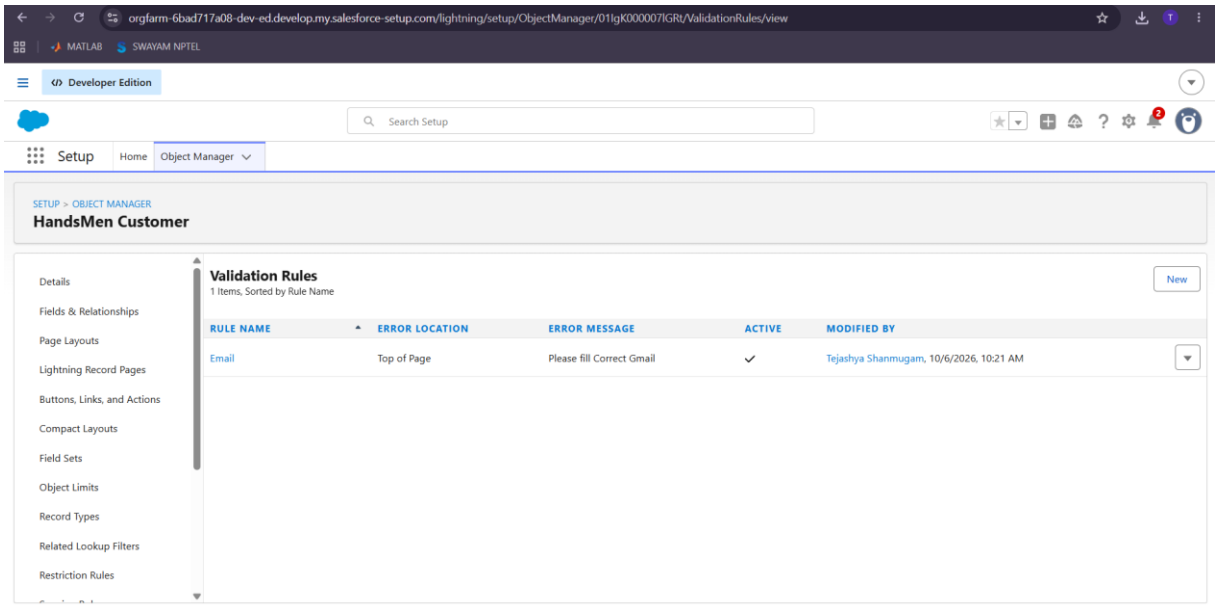


4.4 Validation Rules

Validation rules are used to prevent records containing unacceptable information from being saved. They provide an additional layer of data quality by checking whether entered values satisfy defined business conditions.

When a record fails the configured condition, Salesforce displays an appropriate error message and prevents the invalid data from being committed.

This ensures that important business information remains consistent and reduces errors caused by incorrect user input.

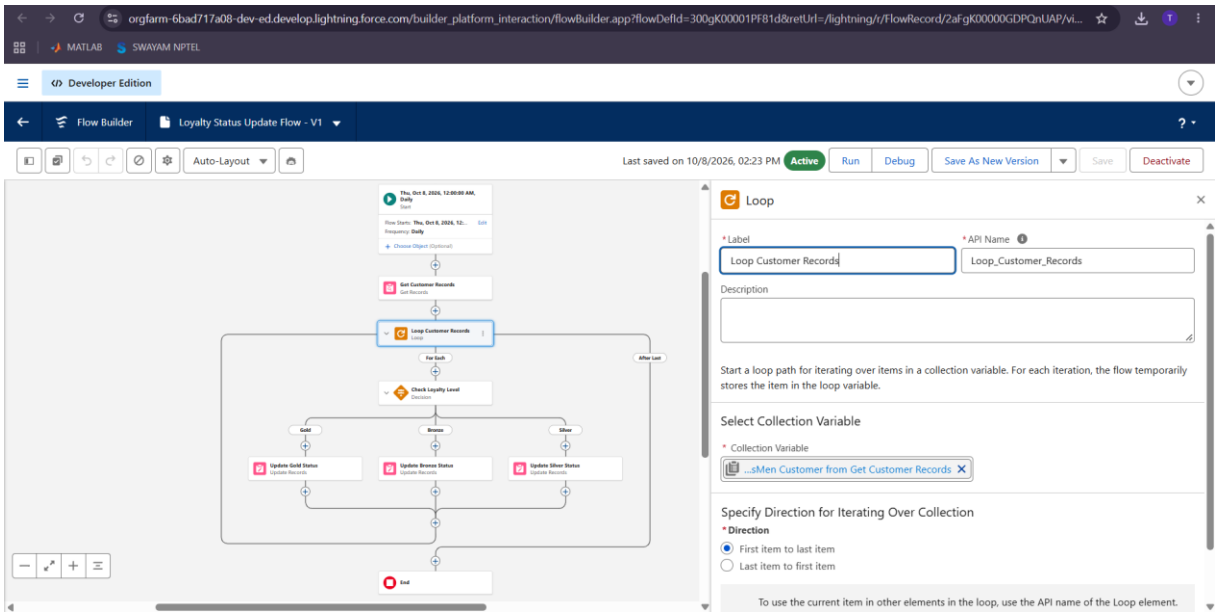


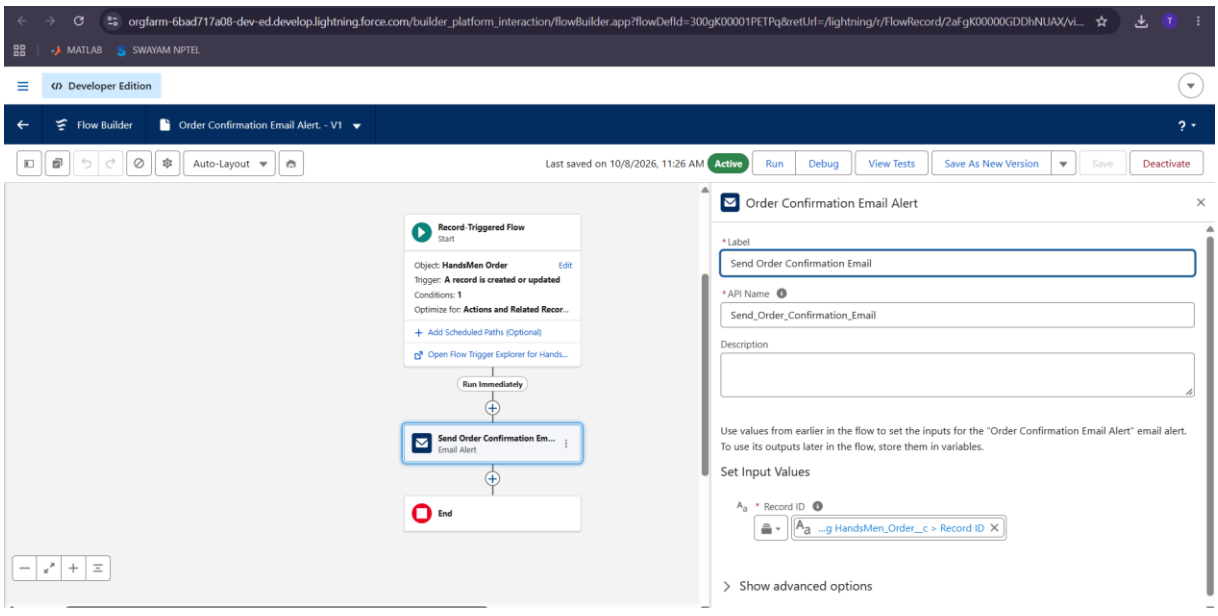
4.5 Automation – Flow

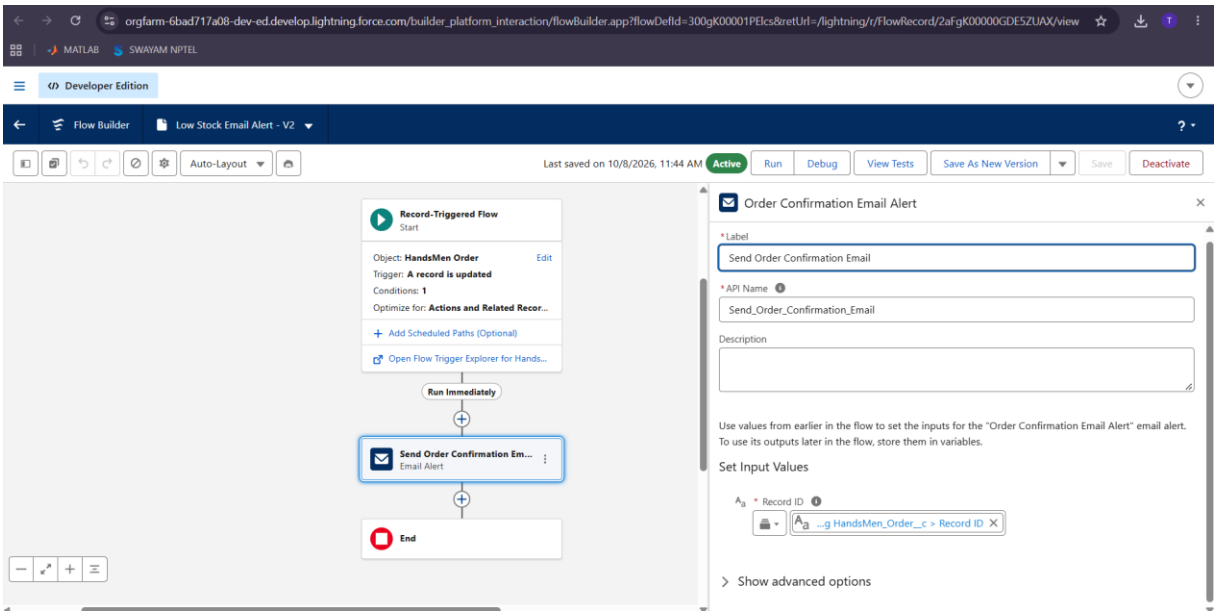
Salesforce Flow is used in the HandsMen Threads CRM to automate the required business process based on predefined conditions. The Flow reduces manual effort by automatically performing the configured action when the triggering condition is satisfied.

The automation improves process consistency and helps ensure that the required business action is completed without depending entirely on manual intervention.

The Flow was tested using sample records, and the resulting action was verified against the expected outcome.







4.6 Approval Process

The Approval Process is used to provide controlled authorization for the relevant business operation. When a record satisfies the configured criteria, it can be submitted for approval and reviewed by the appropriate user.

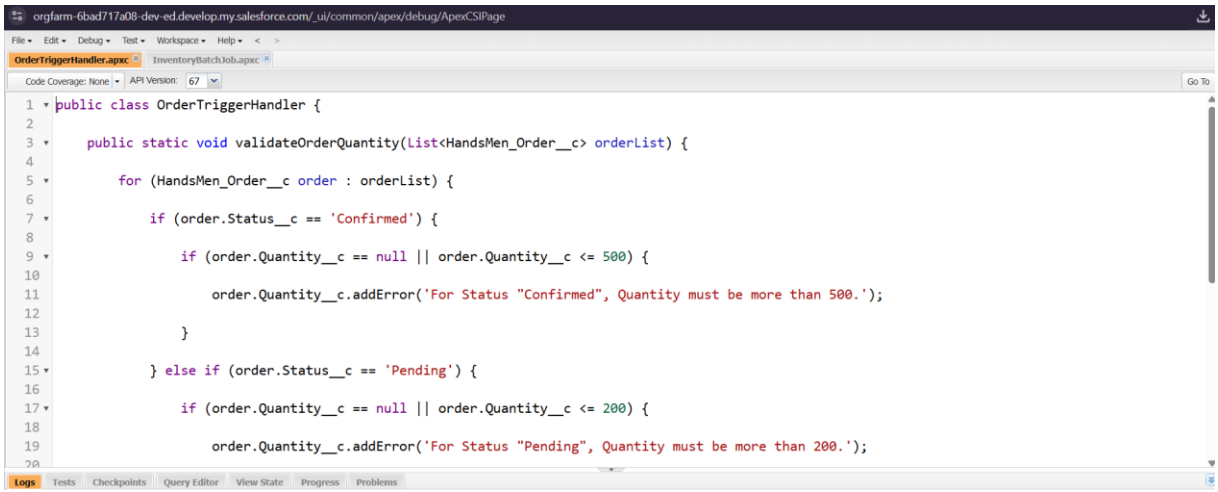
This process helps ensure that important business actions are not completed without the required authorization. The approval workflow was tested by submitting a suitable record and verifying the resulting approval status.

4.7 Apex Classes

Apex Classes are used to implement reusable business logic in Salesforce. In the HandsMen Threads project, Apex classes automate important business processes and improve system efficiency. The Order Trigger Handler class validates order quantities based on their status. For confirmed orders, the quantity must be greater than 500, while pending orders must have a quantity greater than 200. If the conditions are not satisfied, an error message is displayed. The Inventory Batch Job class monitors product inventory and identifies products with a stock quantity below 10. It uses batch processing to handle inventory records efficiently and supports scheduled execution. These classes reduce manual work, maintain data accuracy, enforce business rules, and improve order and inventory management.

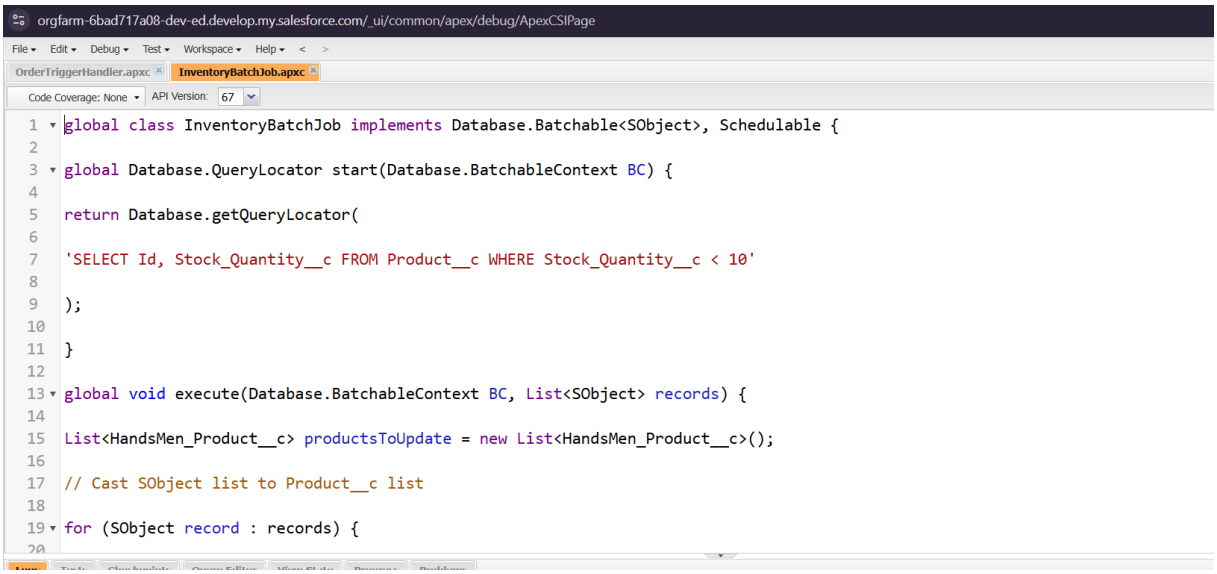
Order Trigger Handler

The Order Trigger Handler Apex class is used to validate order quantities based on their status in the HandsMen Threads project. It processes a list of order records and checks whether the quantity meets the specified business rules. For confirmed orders, the quantity must be greater than 500, whereas pending orders must have a quantity greater than 200. If the quantity is null or does not satisfy these conditions, an error message is displayed, and the record is prevented from being saved. This validation helps enforce business rules, maintain data accuracy, and ensure consistent order processing.



Inventory Batch Job

The InventoryBatchJob Apex class implements the Database.Batchable and Schedulable interfaces to automate inventory monitoring. It retrieves product records from the Product__c object where the stock quantity is less than 10. The batch process handles inventory records in manageable groups, making it suitable for processing large amounts of data. The schedulable interface allows the job to run automatically at configured intervals. This helps identify low-stock products and supports timely inventory management. Overall, these Apex classes reduce manual work, improve data reliability, and enhance the efficiency of order processing and inventory management in the HandsMen Threads application.



4.8 Automation Testing

The automation features were tested using representative records to verify that each component responded correctly to its configured conditions. The testing process involved providing suitable input data, triggering the relevant automation, and checking the resulting system behavior.

The test results were compared with the expected outcomes to confirm that the Flow and Apex triggers performed their intended functions. Input and output screenshots are included as evidence of the testing process.

Test Case Table

Test ID	Feature	Test Scenario	Expected Result	Status
T01	Flow	Valid triggering condition	Flow performs configured action	Pass

5.1 Data Migration

The required project data was added to Salesforce to support testing and demonstrate the working of the HandsMen Threads application. Sample records were created with relevant values so that validations, automation, triggers, reports, and other configured features could be verified using realistic scenarios.

The migrated or created data was reviewed after entry to ensure that the records contained the expected information and could be successfully used by the configured Salesforce components.

5.2 Security and Access Control

Security was implemented to ensure that users receive access according to their responsibilities. Salesforce security features such as Profiles, Permission Sets, Roles, Role Hierarchy, and sharing controls can be used to manage access to objects, records, and system functionality.

The HandsMen Threads project particularly uses **Permission Sets** to provide additional permissions to selected users without changing their existing profile. This approach provides flexibility when different users require different levels of access.

5.3 Testing and Validation

Testing was performed to verify that the major features of the HandsMen Threads application behaved according to their intended requirements. Test records were created for different scenarios, and the actual output was compared with the expected result.

The testing covered important components such as validation rules, Flow automation, stock detection, order quantity validation, reports, and access controls. Successful test results confirmed that the implemented functionality operated correctly under the tested conditions.

6.1 Deployment Strategy

The completed Salesforce configuration and development components can be moved between environments using Salesforce-supported deployment methods such as Change Sets or other appropriate deployment tools. The deployment approach should ensure that required configurations, Apex components, automation, and related dependencies are transferred correctly.

Before deployment, the components should be tested in the development environment to reduce the possibility of errors in the target environment.

6.2 Maintenance and Monitoring

Regular maintenance is important to keep the HandsMen Threads Salesforce application reliable and efficient. The configured automation, Apex triggers, reports, security settings, and data should be periodically reviewed to ensure that they continue to support changing business requirements.

Salesforce monitoring and debugging tools can also be used to identify errors or unexpected behavior. Future modifications should be tested before being introduced into the working application.

Future Enhancements

The HandsMen Threads application can be further enhanced with:

- AI-based sales and inventory predictions
- Customer-support chatbot
- Automated customer notifications
- Advanced inventory forecasting
- Sales forecasting and recommendation features
- E-commerce integration
- Enhanced mobile support
- Advanced analytics and business intelligence

These enhancements could make the system more intelligent and provide better decision-making support as the business grows.

Project Highlights

The major highlights of the HandsMen Threads project are:

- Salesforce-based centralized business management
- Structured data and relationships

- Validation and automation
- **Order Trigger Handler**
- **Inventory Batch Job**
- Permission Set-based access control
- Reports and Dashboard
- Functional testing and validation
- Scalable platform for future AI and analytics integration

Conclusion

HandsMen Threads demonstrates how Salesforce can be customized to support the operational requirements of a fashion and apparel business. The project combines Salesforce configuration, automation, Apex development, reporting, and security features to create a structured CRM solution.

The implementation of validation, automation, stock detection, order processing, reporting, and access control helps reduce repetitive manual work and improve data consistency. The project also provides a strong foundation for future enhancements such as AI-powered analytics, forecasting, and customer-support automation.

Demo Video:

https://drive.google.com/file/d/12ALnQEjjPM2VSAJs_FhMwfS3yJc4n8He/view?usp=drive_link